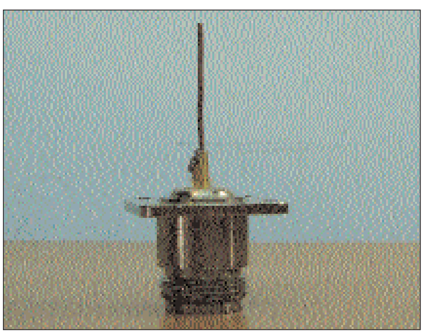
STEP 2 The next step is to prepare the cable. This special cable is called a pigtail. Ready-made pigtails can be expensive, so we'll construct our own. The RG cable that you acquired earlier has two ends (naturally)—one of them should be soldered to the N type female connector. The inner conductor of the coaxial cable will go into the respective protruding metal of the connector while the shield should be soldered to its body. Similarly, solder the other end and then crimp it to the SMA male connector. For the same, first solder the small male pin to the inner conductor of the cable and then insert this into the minute hole though the connector. The pigtail is now ready.


A pigtail from scratch

STEP 3 Take a piece of Gauge 12 copper wire. Solder it to the top of the N type connector.


The soldered copper wire

STEP 4 Now mount the assembly on the can. Let the SMA connector end pass through the hole first. The N type will form a lock and will appear as shown.


The top view

STEP 5 Mark a point on the copper wire such that the arrangement extends approximately to the centre of the can's diameter. Gently pull out the cable and chop off the extra wire above this point.

STEP 6 Put back the cable, and if required, use a few nuts and bolts to secure the assembly. This completes the construction of the reflector.

STEP 7 For the collector bit, attach a nut at one end of the metal rod. Now hold it such that the nut points towards the ground.

STEP 8 Add the lid and slide it to the bottom such that it hugs the nut. Next, place three washers one after the other such that the last one reaches the mid-point of the rod. If the washers don't fit the rod perfectly, try using some household aluminium foil. If there is a large diameter difference, don't lament—get four pieces of a pipe with a larger diameter than the inner rod and washers. Begin by inserting a washer, then a pipe, then a washer, and so on. The pipe would hold all the washers in place since its diameter is greater.


Taming the collector


The final collector

Coming back, insert a roundish insulating material (plastic will do) and let two more washers occupy their respective places on the rod.

STEP 9 The collector can now be mounted. Put the lid back in place with the rod hanging inside the Pringles container. Make sure the rod doesn't touch the copper filament that was previously soldered to the N-type connector.

STEP 10 It's time to test your hack now! Go find a wireless card and a wire-free network or two.

Putting It All Together

If you have a desktop computer with a PCI or USB WLAN card, you're probably already testing your newfound passion! Unfortunately, if you are the lucky owner of a laptop with an inbuilt Wi-Fi system, life isn't that easy. You will have to open a small cover, generally found on the lower surface of the laptop. This will expose a couple of cables running up to the inbuilt antenna. Think twice before pulling them out and inserting your own—make sure you don't end up voiding your warranty in the process. However, if you know what you are doing and do it correctly, there should not be any problems.

Testing Times

Yagi antennas (like the one you just made) are highly directional, and this signifies greater range (remember the highly directional laser light?). Also, noise is kept down for neighbours who share the same channel as you.

To establish a connection with someone, start by aiming slightly towards the left or right of their receiver, and slowly rotate the assembly to find the best gain possible. Elevating the antenna can sometimes help.

If the antenna is to be placed amidst open surroundings, some weatherising needs doing—paint the external surface to ward off rust, or enclose the whole assembly in a PVC pipe.

Wrapping It Up

We have discussed the basic antenna, and it is from here that the true hacks emerge. You have learnt how to construct a relatively tough-to-build antenna, and a tin-can setup should follow similar but easier steps. So put your creative shoes on—and don't forget to aim at the right building! ☒

readersletters@jasubhai.com